Ad Omnia

Ad Omnia in perpetuum.



universidade de aveiro
theoria poiesis praxis

instituto de
telecomunicações

"Knowledge is Power"

- *Frequently attributed to Sir Francis Bacon*

Requirements Gathering

Meetings with entities such as:



Commander of Voluntary Firefighters of Ílhavo
Artur Ferreira



Sargento Chefe GNR
César Paradela

Personas



Sofia Marques

Control Room Operator



Tiago Ferreira

Emergency Planner



Ana Costa

Citizen

PERSONA CV · AD OMNIA

Control Room Operator



NAME
Sofia Marques

AGE
34

ROLE
Shift operator in the control room of a sub-regional emergency and civil protection command

APPLICATION
Private application (control room)

● ● ● **Tech proficiency — High.** Works with several systems simultaneously during long shifts.

BACKGROUND

Works shifts in the control room: monitors occurrences, keeps incident records current, tracks resources, and is the main point of contact with other agencies and field teams during a crisis.

GOALS

- Get a complete, up-to-date picture at a glance, with minimal manual work.
- Anticipate how an incident will evolve, not just react to it.
- Allocate resources quickly and justify decisions afterwards.
- Quickly find information from other agencies.

FRUSTRATIONS

- Information spread between different systems.
- Not knowing whether the data she sees is current or outdated.
- Information from agencies arriving late, incomplete or contradictory.
- No record of why a decision was made, when questioned later.

WHAT SHE NEEDS FROM AD OMNIA

- A Common Operational Picture, all agencies on one map.
- Automatic ingestion of external sources, last-updated shown.
- Create, update and close incidents; track resources and orders.
- Alerts on new incidents or relevant changes.
- Incident progression predictions, with confidence shown.
- Resource and routing recommendations she can accept or reject, logged.
- A way to publish incidents and warnings to the public.

PERSONA CV · AD OMNIA

Emergency Analyst



NAME
Tiago Ferreira

AGE
41

ROLE
Analyst at the sub-regional Aveiro command

APPLICATION
Private application (control room)

●●● **Tech proficiency — High.** Uses GIS tools and spreadsheets regularly.

BACKGROUND

Works mostly outside active emergencies: prepares the region for wildfire season, plans safety arrangements for large events, and analyses past occurrences to identify high-risk areas.

GOALS

- Identify high-risk areas and periods based on historical data.
- Simulate scenarios before large events or critical seasons.
- Produce reports that support planning and resource requests.

FRUSTRATIONS

- Historical data is scattered and in inconsistent formats.
- Plans are based on experience and intuition rather than data.
- No way to test a plan before it is needed.

WHAT HE NEEDS FROM AD OMNIA

- Access to historical incident data, filterable by area, period, type.
- Scenario simulation using the predictive engine and synthetic data.
- Export of maps, data and reports.
- Test whether current resources are enough under extreme demand.

PERSONA CV · AD OMNIA

Citizen



NAME
Ana Costa

AGE
38

ROLE
Teacher, living in a rural area close to forest

APPLICATION
Public application

Tech proficiency — Medium. Uses her smartphone daily, does not install many apps.

BACKGROUND

Lives with her family in a village surrounded by forest. Every summer she follows wildfire news anxiously, often relying on social media posts of uncertain origin.

GOALS

- Know whether there is an active emergency near home, family or route.

FRUSTRATIONS

- Contradictory or alarmist information on social media.
- Official information hard to find or not specific to her area.

WHAT SHE NEEDS FROM AD OMNIA

- A simple map of active emergencies, not limited to wildfires.

Identity Guidelines:
“A Thousand No’s”

No.



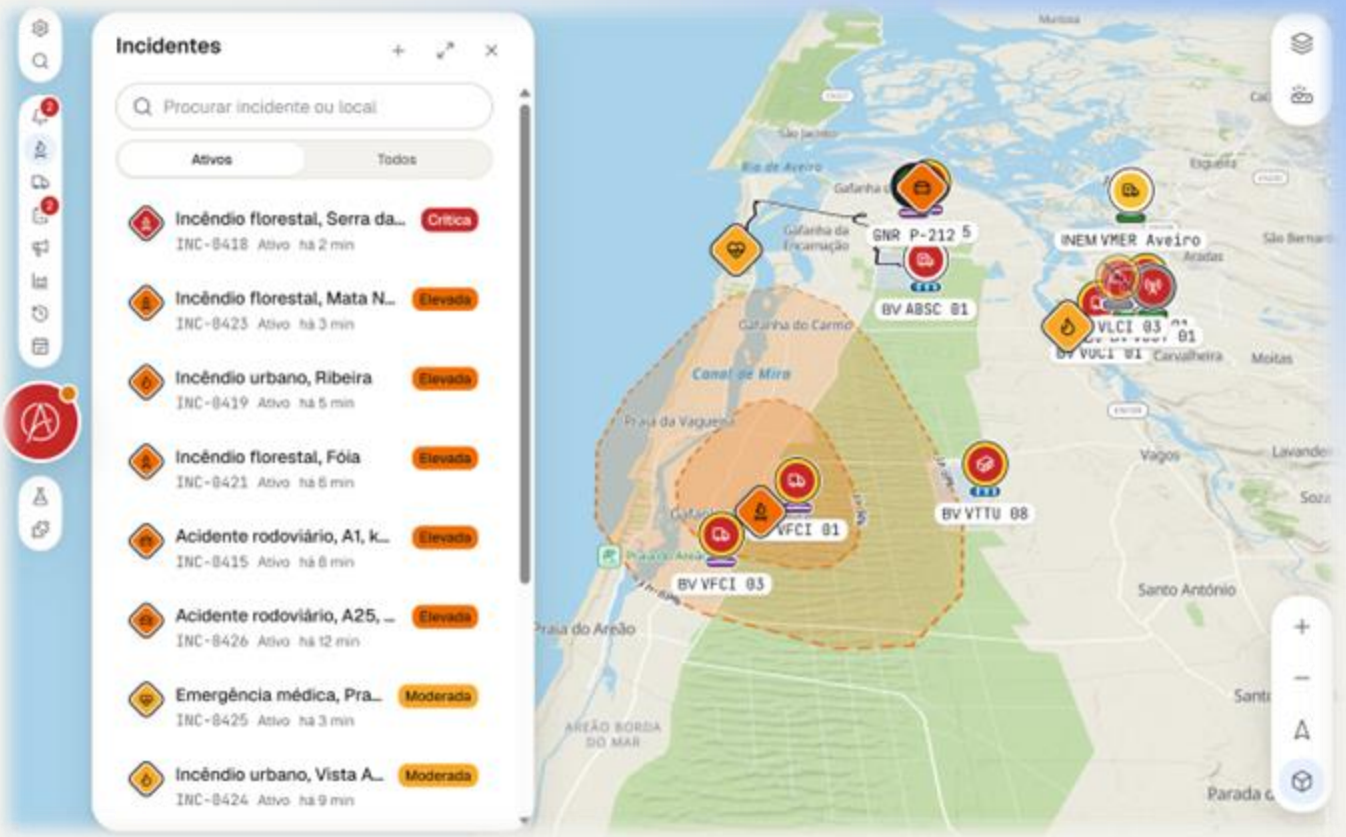
Stretched logo

User Stories – US-1

As... a control room operator,

I want... to view on a single real-time map all active incidents and the location and status of resources from every agency, including aerial assets,

So that... I have a complete and up-to-date picture of the situation without switching between multiple systems.

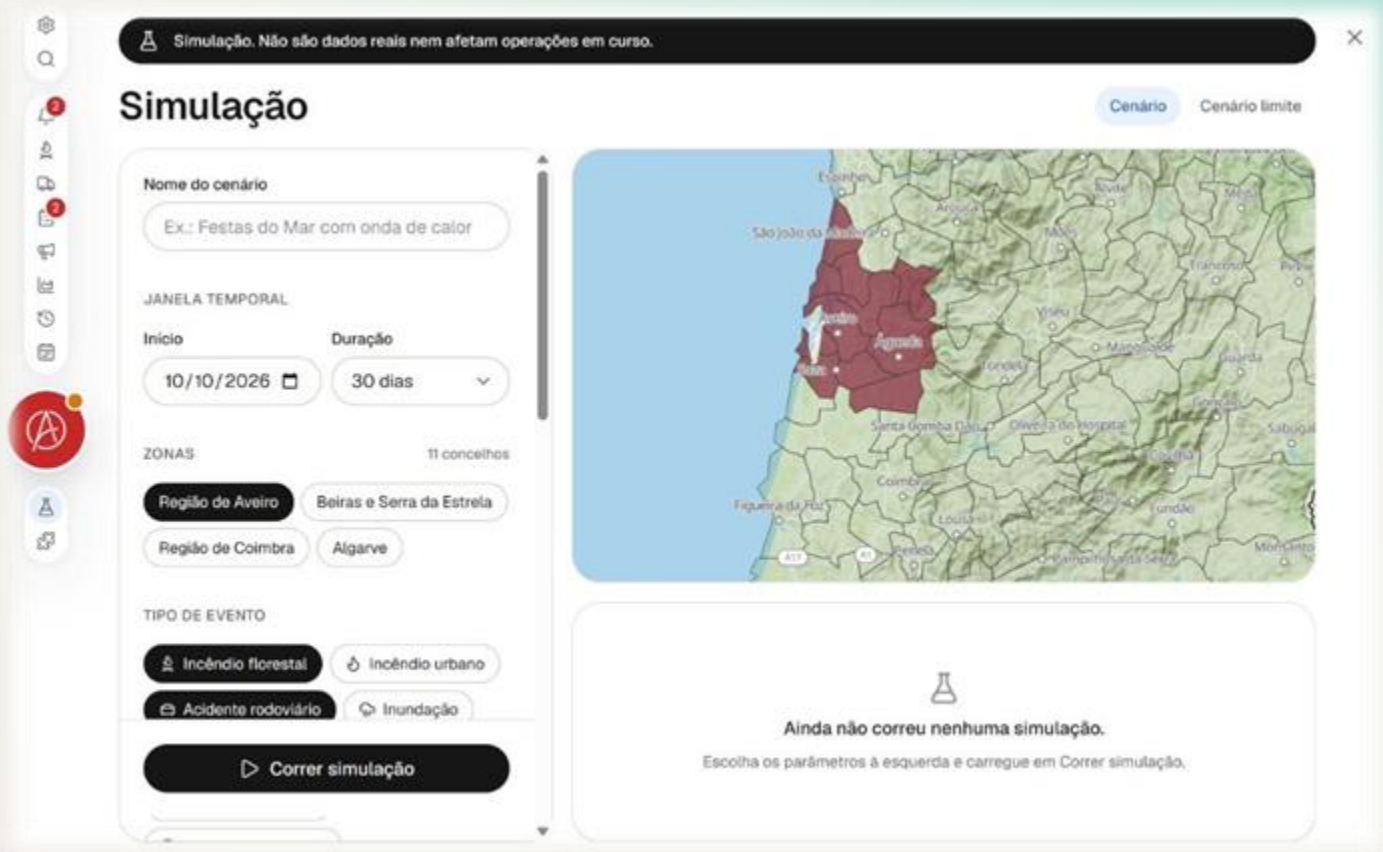


User Stories – US-12

As... an analyst,

I want... to run simulations using the predictive engine and synthetic data before major events or the start of fire season,

So that... I can anticipate likely demand and prepare resource allocation in advance as well as requesting help timely.

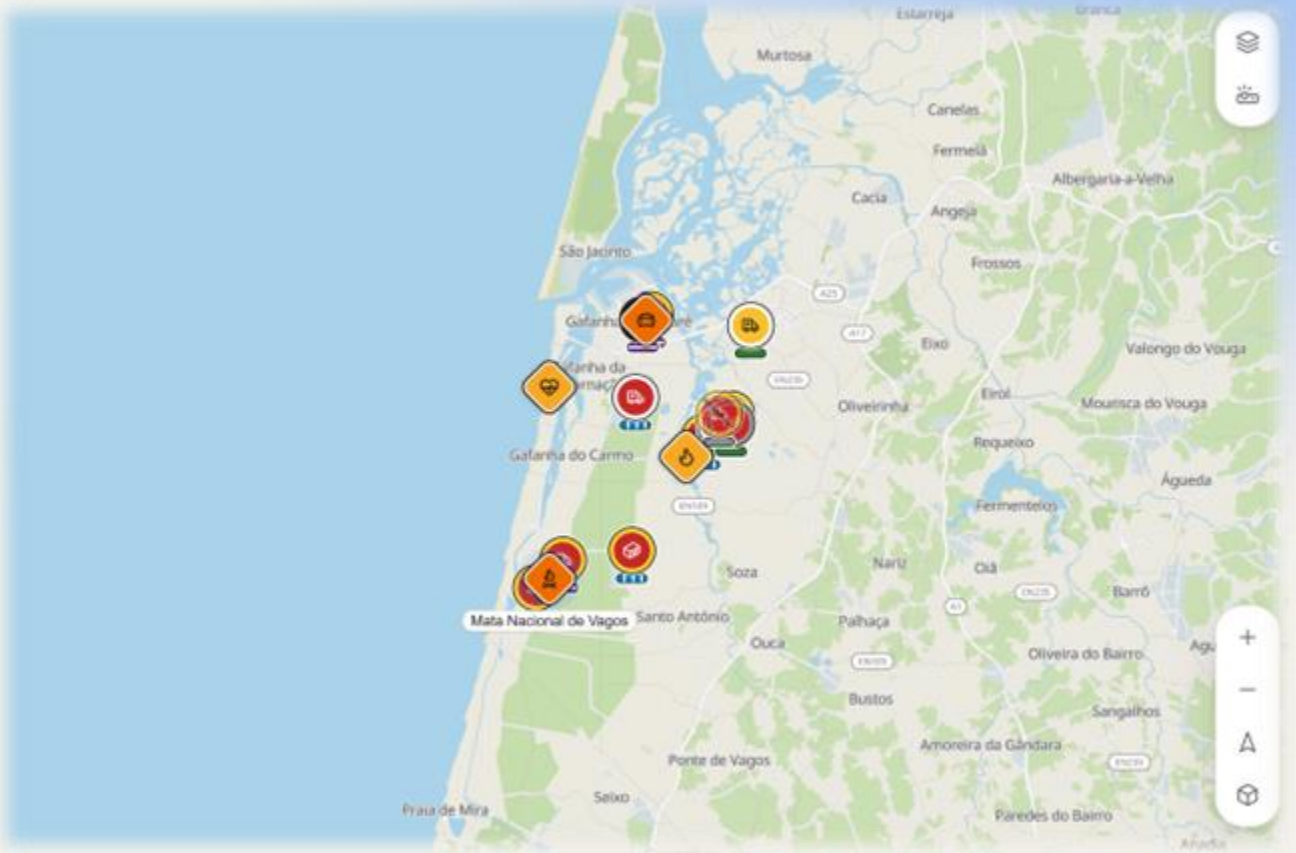


User Stories – US-15

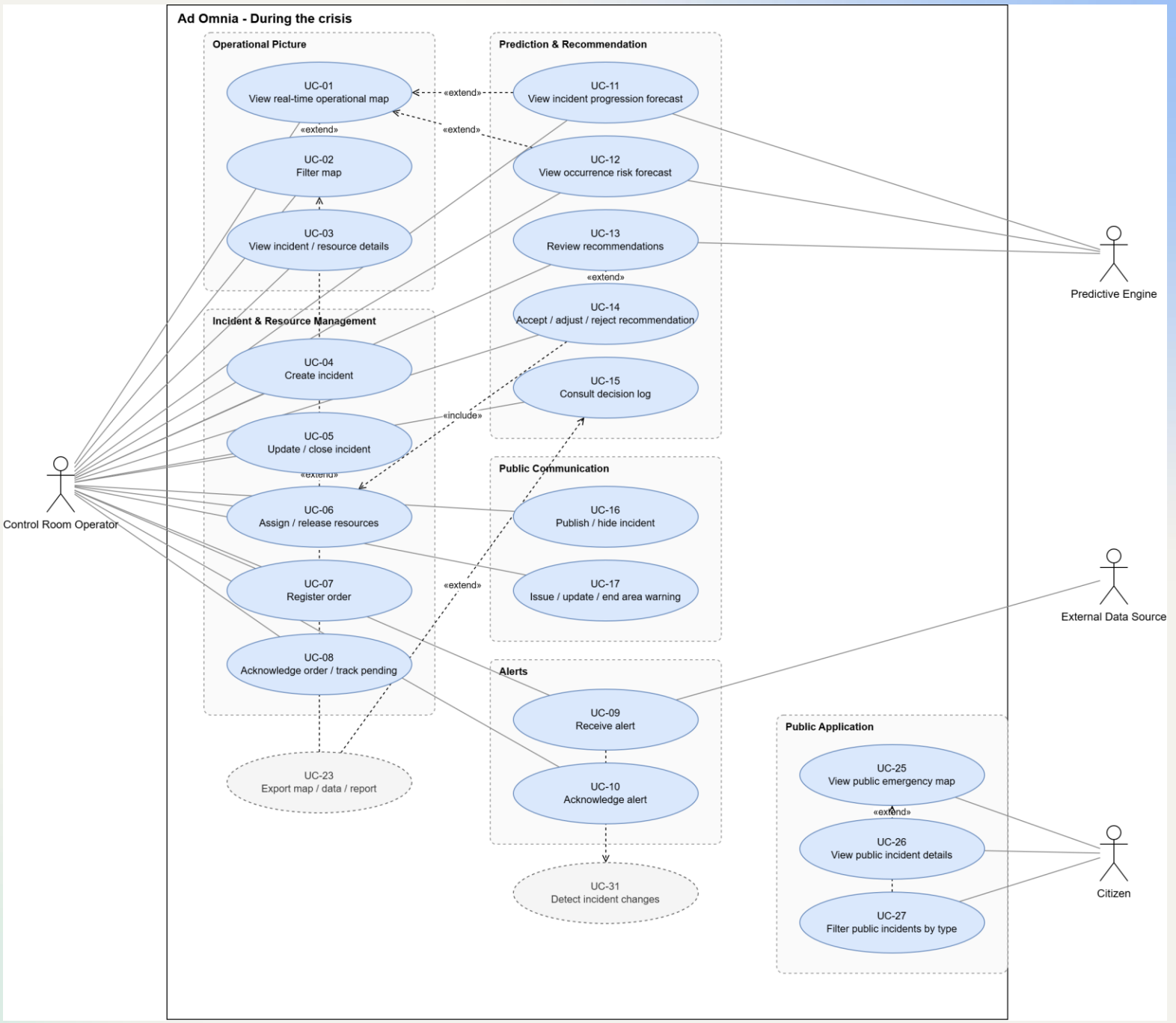
As... a citizen,

I want... to see the active emergencies of any type on a simple map,

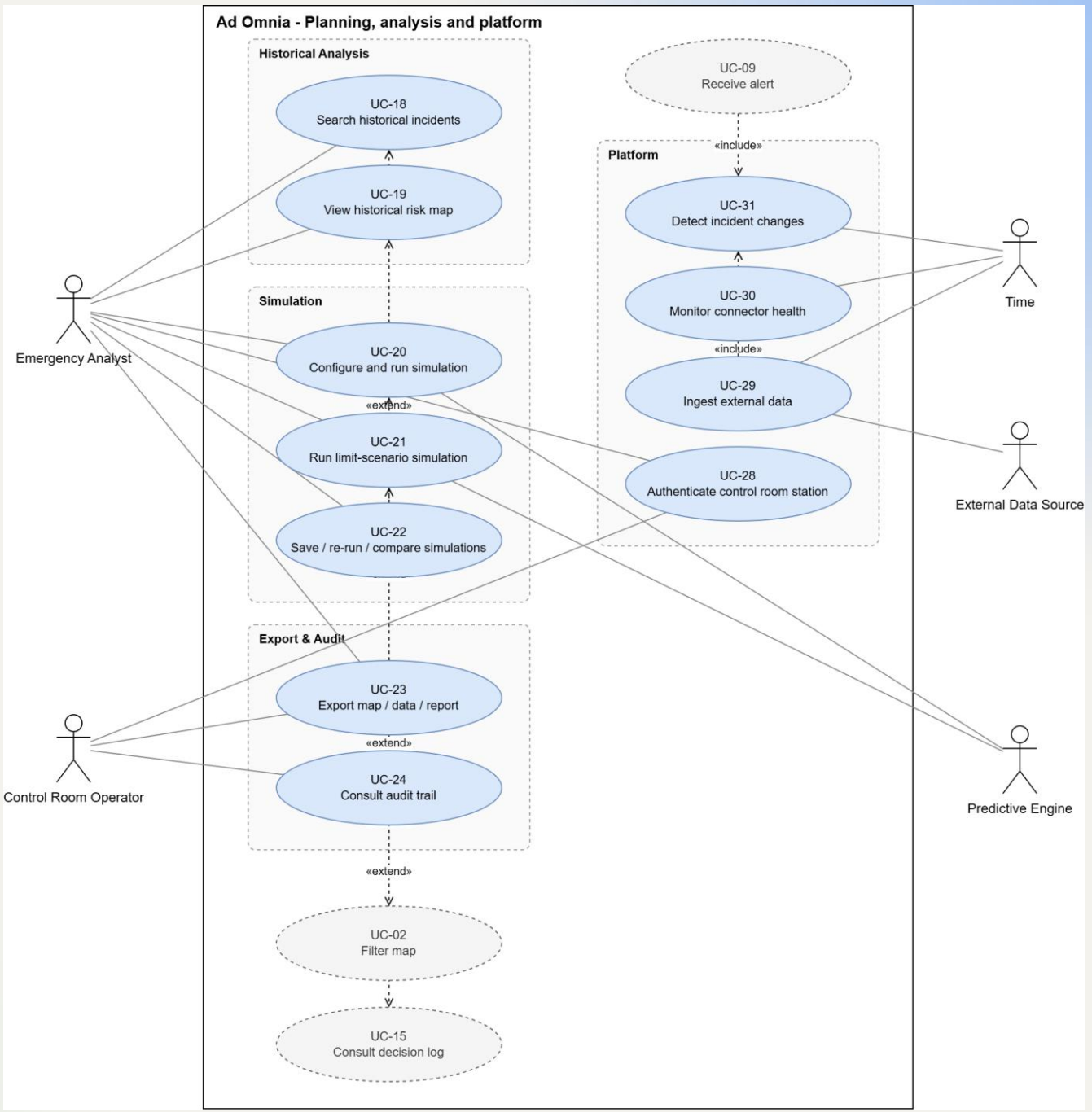
So that... I can quickly tell whether there is an emergency near my home, my family or my route, without relying on social media or unofficial sources.



Use Cases – During the Crisis



Use Cases – Before and After the Crisis



Functional Requirements

Data Ingestion (ING)	3 items
Operational Picture (COP)	2 items
Historical analysis (HIST)	3 items
Incident and resource management (INC)	3 items
Alerts (ALR)	1 item
Prediction (PRD)	5 items
Recommendation (REC)	3 items
Audit (AUD)	2 items
Export and reporting (EXP)	1 item
Public publishing (PUB)	8 items
Total	**31 items**

Non-Functional Requirements

Performance (PERF)	2 items
Availability (AVL)	4 items
Integrity (INT)	1 item
Security (SEC)	3 items
Scalability (SCA)	1 item
Usability (USA)	2 items
Total	**13 items**



Assumptions & Dependencies

- This project is a proof of concept rather than a final solution.
- Some data is classified and some APIs are not public.
- Predictive AI module can run on synthetic/simulated data.
- Field responders stay off-platform (orders reach them outside the app)
- External source APIs (Fogos.pt, ADS-B) - breaking changes or access loss would block those plugins.

"To create, one must first question everything."

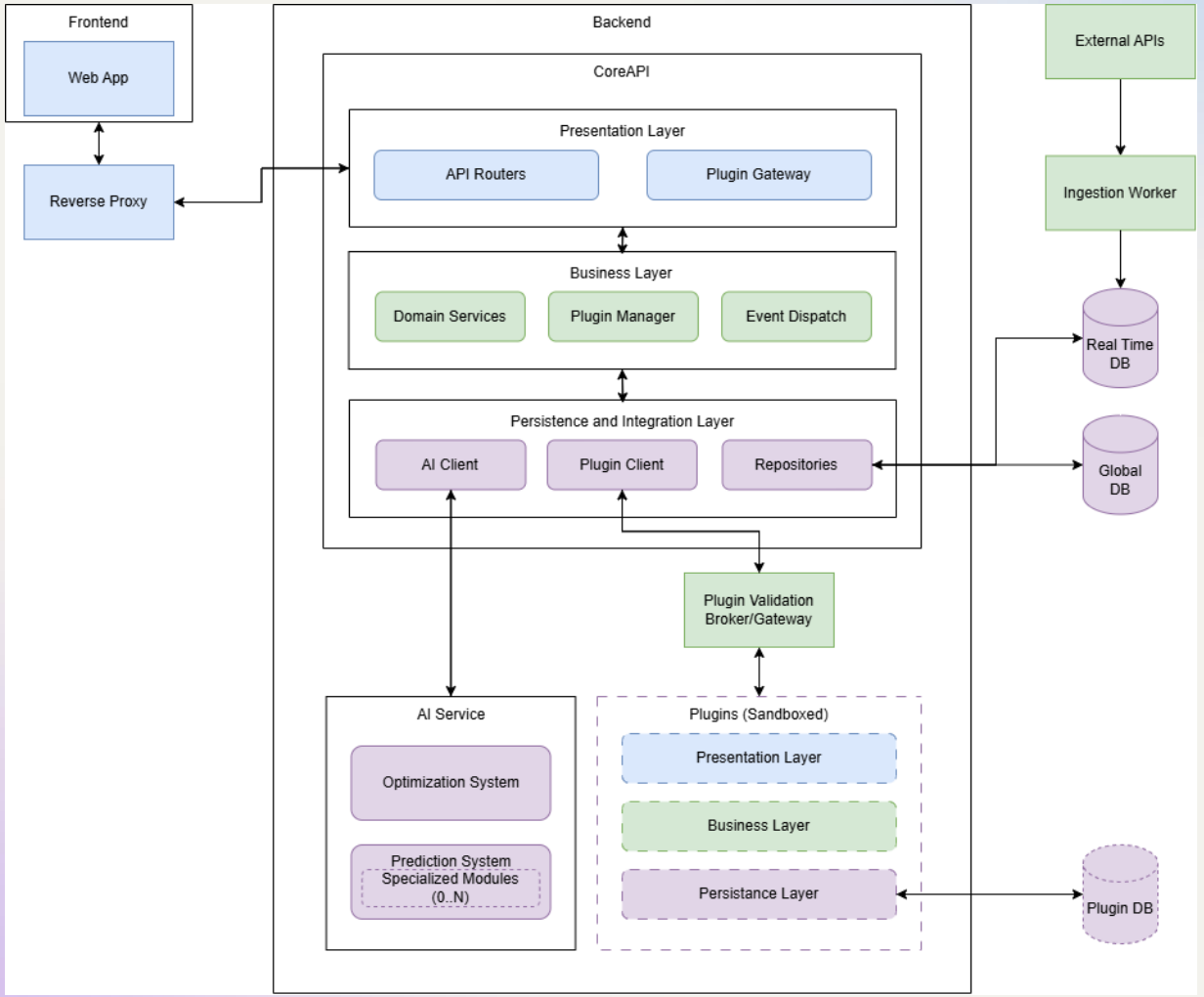
- *Eileen Gray*



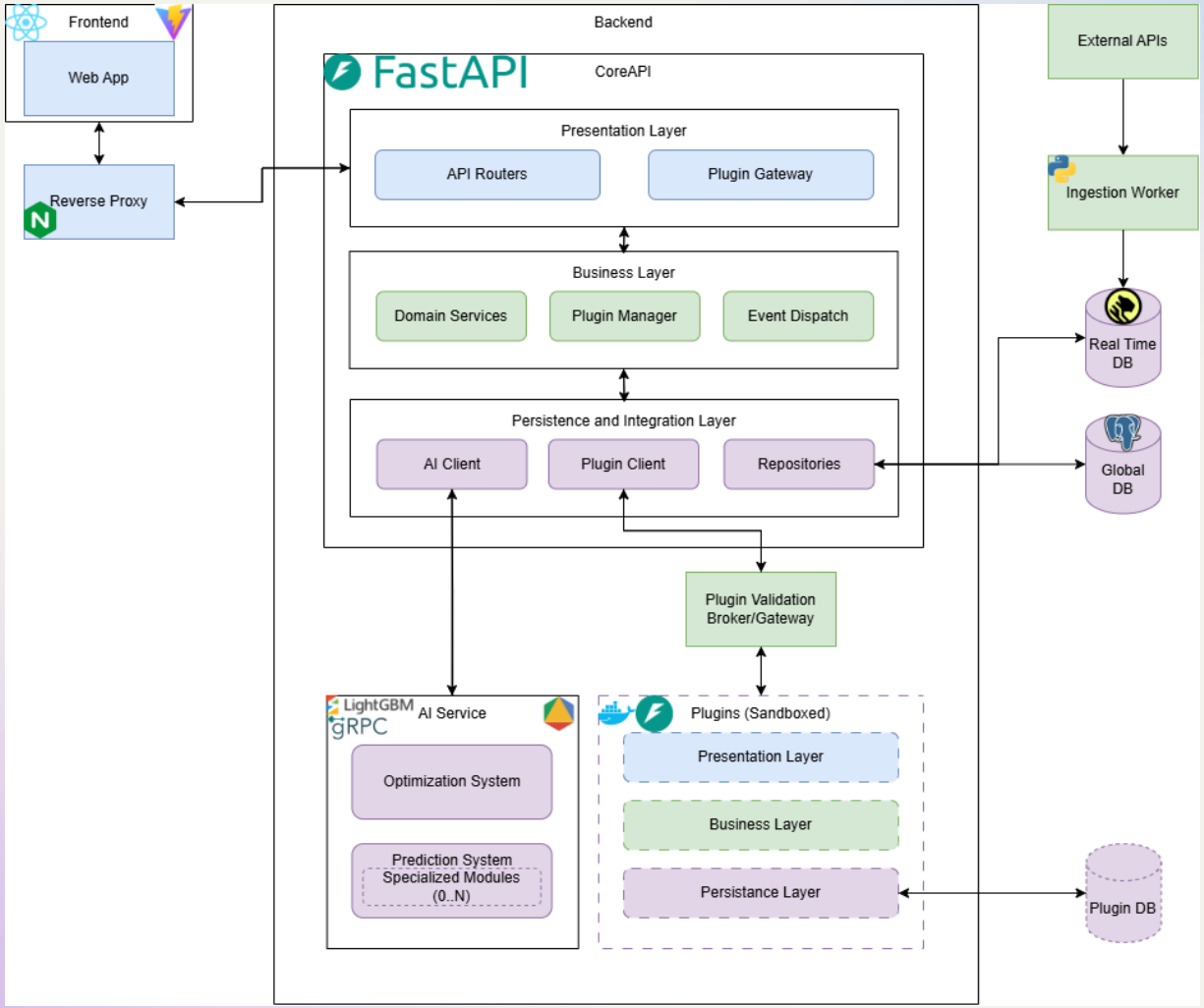
Updated State-of-Art

	Feature	Palantir Gotham	Thales Hexforce	SADO	Octave OnCall	Esri ArcGIS	**Ad Omnia**
Operator	Decision Logging and Auditability	●	◐	○	●	◐	●
	Visual AI Prediction with Confidence Scores	◐	◐	○	◐	●	●
	Integrated Public Warning Publishing	○	○	◐	◐	◐	●
Analyst	Historic Multi-Filter Incident Analytics	●	◐	◐	●	●	●
	Synthetic Data and Scenario Simulation Engine	◐	◐	○	○	◐	●
	Extreme Demand & Resource Stress Testing	◐	○	○	○	◐	●
	Automated Map, Data & Report Exporting	●	●	○	●	●	●
Cit.	Multi-Hazard Public Emergency Map	○	○	◐	◐	●	●

System Architecture



Tech Stack


Frontend
Web App
Reverse Proxy
Backend
FastAPI
CoreAPI
Presentation Layer
API Routers
Plugin Gateway
Business Layer
Domain Services
Plugin Manager
Event Dispatch
Persistence and Integration Layer
AI Client
Plugin Client
Repositories
Plugin Validation Broker/Gateway
LightGBM gRPC
AI Service
Optimization System
Prediction System Specialized Modules (0..N)
Plugins (Sandboxed)
Presentation Layer
Business Layer
Persistance Layer
External APIs
Ingestion Worker
Real Time DB
Global DB
Plugin DB

Ad Omnia in perpetuum.

To everything, forever.

